Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	11
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2088.2	1801	1031.95	px^2
major_axis_length_px	lacuna shape	85.9334	87.8115	23.7232	px
minor_axis_length_px	lacuna shape	31.5731	30.3414	8.451	px
aspect_ratio	lacuna shape	2.8283	2.7498	0.7793	unitless
eccentricity	lacuna shape	0.9147	0.9291	0.0652	unitless
solidity	lacuna shape	0.8718	0.8885	0.0596	unitless
roots_count	headline	9.5455	8	2.9787	count
ring_length_r30_px	headline	362.909	385	89.6286	px
ring_length_r60_px	reported	914.545	925	178.007	px
owned_length_px	ownership-dependent	2779.71	2005.65	1602.73	px
edge_count	ownership-dependent	132.364	104	83.8824	count
mean_edge_length_px	ownership-dependent	21.7334	21.3532	2.6314	px
perimeter_px	normalised	207.522	210.167	71.7085	px
ring_area_r30_px2	normalised	8536.09	8646	1979.1	px^2
ring_area_r60_px2	normalised	21938.7	22056	3981.31	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0429634	0.0413851	0.00778076	px^-1
ring_density_r60	normalised	0.0421177	0.0403625	0.00688033	px^-1
roots_per_100px_perimeter	normalised	5.1793	4.3052	2.6638	per 100 px
ring_attached_length_r30_px	network v2	324.818	316	80.9862	px
ring_attached_length_r60_px	network v2	719.545	633	166.341	px
ring_length_w_r30_px	network v2	408.738	428.463	98.0826	px
ring_length_w_r60_px	network v2	1032.18	1043.36	191.231	px
sholl_crossings_r10	network v2	8.7273	9	2.7236	count
sholl_crossings_r20	network v2	11.0909	12	2.9139	count
sholl_crossings_r30	network v2	12.8182	14	3.6556	count
loop_count	loops, report only	13.7273	10	12.4747	count
loop_edge_share	loops, report only	0.345	0.3945	0.1311	share
ring_width_mean_r30_px	width	7.2057	7.2449	0.3751	px

Per field
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measure	value	units
analysed_area_px2	1.0246e+06	px^2
total_skeleton_length_px	37067	px
canalicular_length_density_per_px	0.0361772	px^-1
junction_count	864	count
junction_density_per_px2	0.00084326	px^-2
lacuna_count	12	count

lacunae_per_px2	1.171e-05	px^-2
skeleton_component_count	255	count
mean_component_length_px	145.361	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0409714	px^-1
field_density_without_flagged_per_px	0.0372077	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	560	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0361772	px^-1
field_length_density_with_unattached_per_px	0.0367238	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	177	count
loop_edge_share	0.3585	share
edge_count	1911	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8093	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	35744	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
2	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
3	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
4	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
5	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
6	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
7	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
8	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
9	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
10	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
11	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False
12								180.095	840.839	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	386	840	2005.65	104	19.2851	7.2461
2	15	470	1115	5039.86	256	19.6869	7.6645
3	6	298	790	2560.8	122	20.9902	7.8485
4	10	392	925	4122.75	192	21.4727	7.2449
5	10	460	980	1776.43	92	19.309	7.3791

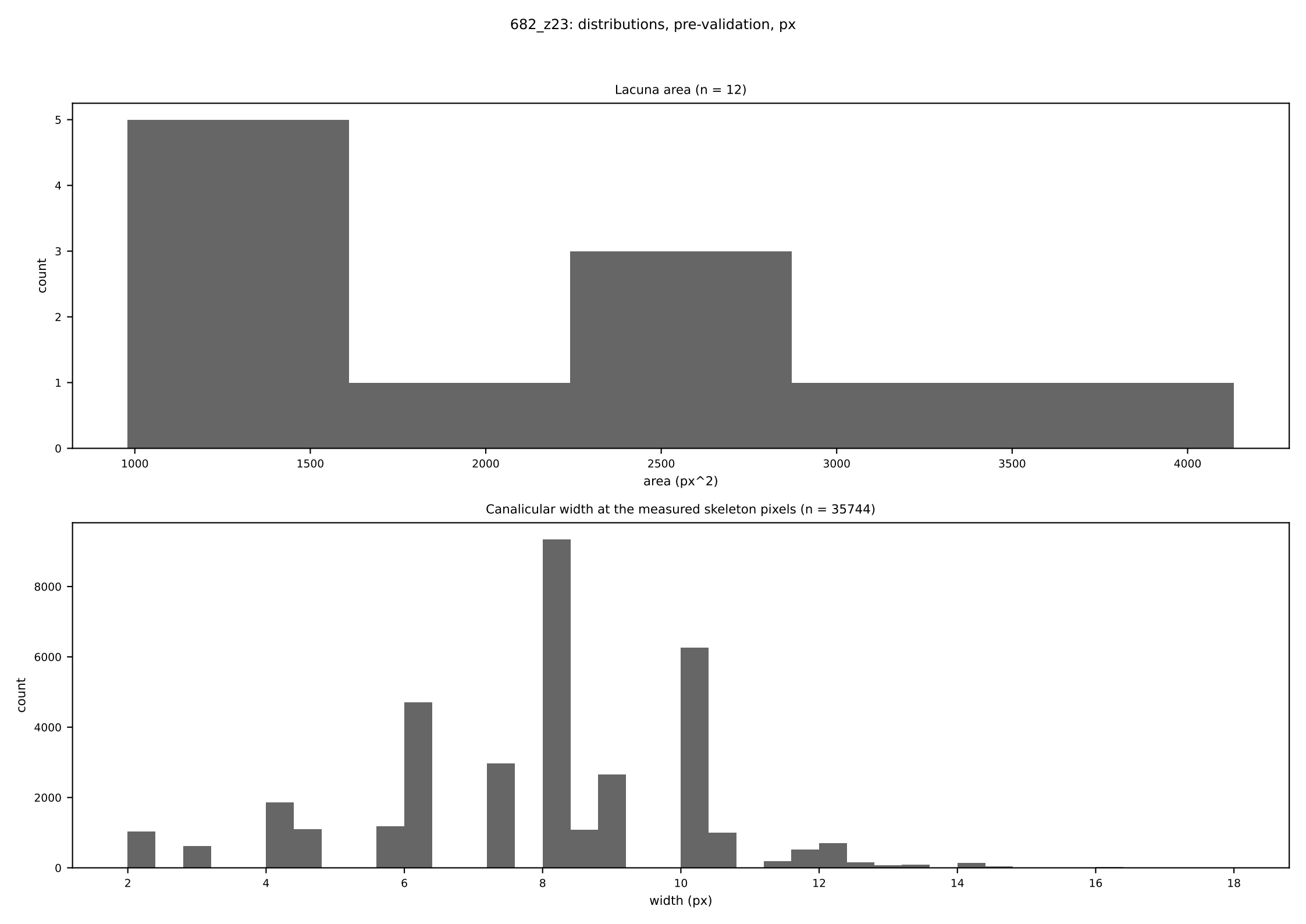
6	10	351	815	1273.48	53	24.0279	7.4606
7	8	471	1274	3081.6	135	22.8266	7.3243
8	7	289	706	1835.23	66	27.8065	6.9399
9	14	385	954	1387.96	65	21.3532	6.4529
10	8	255	793	1453.53	67	21.6944	7.1906
11	12	379	1002	1321.46	53	24.9332	6.9559
12	7	207	681	5991.6	304	19.7092	7.016

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	219.681	8976	22806	1	0.9689	0.0430036	0.0368324	3.6416
2	335.421	12178	29932	1	1	0.0385942	0.0372511	4.472
3	167.238	7485	20582	1	1	0.039813	0.0383831	3.5877
4	243.522	9472	22056	1	1	0.0413851	0.0419387	4.1064
5	167.924	7349	17555	1	1	0.0625935	0.0558246	5.9551
6	282.894	8678	19332	0.813	0.7462	0.0404471	0.0421581	3.5349
7	263.48	10172	24552	1	1	0.0463036	0.0518899	3.0363
8	174.51	7570	20770	1	1	0.038177	0.0339913	4.0112
9	210.167	8646	22189	1	1	0.0445293	0.0429943	6.6614
10	185.823	7980	21610	1	1	0.0319549	0.036696	4.3052
11	259.279	9669	24825	1	1	0.0391974	0.0403625	4.6282
12	55.6985	4400	14449	1	1	0.0470455	0.0471313	12.5677

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	300	526	428.463	928.068	10	12	14
2	432	790	518.505	1237.13	10	14	14
3	281	622	346.635	913.193	9	9	8
4	316	626	439.534	1043.36	9	13	12
5	448	951	520.162	1099.67	11	13	15
6	259	547	400.191	919.21	9	12	11
7	416	1062	525.777	1428.7	12	14	19
8	248	572	327.179	815.009	6	11	10
9	327	768	443.961	1077.99	9	11	17
10	253	609	289.693	907.293	8	9	11
11	354	756	422.392	1123.51	10	12	14
12	198	633	233.823	780.027	2	4	7


682_z23: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1000
1500
2000
2500
3000
3500
4000
Canalicular width at the measured skeleton pixels (n = 35744)
count
width (px)
0
2000
4000
6000
8000
2
4
6
8
10
12
14
16
18